# JOHAN DAVID RODRIGUEZ CASTRO

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## PROFESSIONAL PROFILE

Final-semester Systems and Software Engineering student (UPB Bucaramanga; expected to graduate in 2026) working at the intersection of security engineering and applied machine learning. I ship tools rather than findings: an open-source supply-chain scanner, a phishing classifier that audited its own benchmark and published the honest number instead of the flattering one, and a vulnerability prioritisation dashboard that measured away the advantage it was built to demonstrate. Equally comfortable in cloud infrastructure — AWS, Terraform, CI/CD — and full-stack delivery, with Scrum Master experience on an academic project and verified C2 English (EF SET). Seeking an AI/ML (model training and evaluation) or Cybersecurity role.

## TECHNICAL SKILLS

**Security & Detection:** Software supply-chain security (slopsquatting), detection-as-code, MITRE ATT&CK mapping, risk-based vulnerability prioritisation (CISA KEV / EPSS / CVSS), phishing and URL analysis, TOTP 2FA, secure SDLC and CI security gates

**Data, ML & AI:** LightGBM, XGBoost, scikit-learn; data-leakage detection and honest evaluation design; SHAP explainability; grouped, temporal and walk-forward validation; probability calibration; FinBERT NLP; CrewAI multi-agent orchestration; RAG with ChromaDB; pandas / NumPy

**Cloud & DevOps:** AWS (IAM, VPC, EC2, Auto Scaling, ALB, CloudWatch), Terraform (IaC), Docker, GitHub Actions CI/CD, k6, Google Cloud (BigQuery ML, Cloud Storage, DLP)

**Programming:** Python, TypeScript / JavaScript, SQL, Dart, HCL, LaTeX

**Databases:** PostgreSQL, MySQL, SQLite, DuckDB, ChromaDB; hybrid SQL / NoSQL architectures

**Methods & Tools:** Agile / Scrum, spec-driven development, pytest, Playwright, mypy strict, UML modelling, Git

## EXPERIENCE

**Developer / Technical Support (Volunteer)** — Jan 2026 – Present | Pontificia Universidad Bolivariana

GOATGuard client: github.com/Yoyagm/goatguard-app

- Technical support for the institution's end users.
- Endpoint threat detection: identifying and analysing port scans and network intrusion attempts.
- Automatic classification of network assets (routers, printers, cameras) with a real-time monitoring dashboard and KPIs.
- GOATGuard Android client (Flutter), a team network-monitoring project: real-time dashboard, inventory, severity-filtered alerts, TOTP 2FA and the JWT kept in OS-backed secure storage. Author of 19 of the client repository's 25 commits.

## KEY TECHNICAL PROJECTS

**SlopGuard** — *Open-Source Supply-Chain Security Scanner* — 2026 | Independent

github.com/Yoyagm/slopguard

- Built a pre-install guard that scores PyPI and npm dependencies against hallucinated, typosquatted and malicious names *before* any code executes, using a five-layer detection engine with zero runtime dependencies.
- Designed a provable anti-false-positive invariant: the opt-in LLM hallucination layer is structurally unable to block a legitimate package on its own.
- Enforced correctness with a CI gate at $\geq$90 % global coverage ($\geq$95 % on critical paths) and eight import-linter architecture contracts, and shipped four integration points from one core: CLI, pre-commit hook, GitHub Action and self-hostable SaaS.

**Phishing URL Detector** — *Explainable Detection & Dataset Leakage Audit* — 2026 | Independent

yoyagm.github.io/phishing-url-detector | github.com/Yoyagm/phishing-url-detector

- Audited PhiUSIIL (UCI id 967, 235,795 URLs) before training and measured two independent label leaks: a one-line rule scores 99.67 % accuracy, and hand-rolled lexical features on the raw URL — the obvious fix — still score 99.51 % while measuring nothing.
- Retrained on canonicalised host names only and published the honest result: ROC-AUC 0.937 and 70.5 % recall at a 1 % false-positive rate, against phishing submitted two years after the training data was collected.
- Reimplemented exact path-dependent TreeSHAP in dependency-free JavaScript, parity-tested at 1e-12 against the Python reference over 28,917 attributions, so the live demo explains every verdict with no backend.
- Delivered 113 tests — including leakage guards that ship with planted-leak negative controls — a containerised FastAPI service and a nine-page technical report in LaTeX.

**Vulnerability Prioritisation Dashboard** — *CISA KEV + EPSS + NVD* — 2026 | Independent

github.com/Yoyagm/vuln-prioritization-dashboard

- Built a reproducible ingestion pipeline over three public feeds ranking 359,399 live CVEs, with committed data contracts and schema tests that fail the build when a feed changes shape.

- Reproduced the industry claim that EPSS is 6.1× more efficient than CVSS, then showed — with scores frozen at five past cutoffs and only later-confirmed exploitation counted — that most of it is evaluation artefact: EPSS wins the head of the queue (1.2–1.8×) and loses the long tail (0.5–0.7×).
- Bootstrapped 95 % confidence intervals over the positives and guarded the tie-averaging in the core metric with a test that ships with a negative control; 76 tests and five ADRs.

**AWS High-Availability Infrastructure** — *Terraform, FastAPI & CI/CD* 2026 | UPB (course final assessment)
github.com/Yoyagm/eval-final-dgiti

- Codified a three-tier architecture across two availability zones in Terraform: VPC, ALB, Auto Scaling Group (minimum 2 instances), PostgreSQL 15 primary plus streaming-replication replica on EC2 in private subnets (AWS Academy does not allow RDS), and CloudWatch alarms.
- Enforced least privilege with a security-group chain — the database has no public IP and accepts connections only from the application tier; its outbound access is open for image pulls.
- Automated deployment with three GitHub Actions workflows (CI; CD with image push to GHCR and ASG instance refresh; Terraform plan on pull requests), keeping Terraform out of the hot path, and validated load balancing under k6 load against CloudWatch metrics.

**Autonomous Market Intelligence System** — *Applied ML & Multi-Agent AI* 2026 | Independent
Private repository — walkthrough available on request

- Built an event-driven pipeline integrating market-data APIs, technical-indicator computation and a FinBERT sentiment model, orchestrating four CrewAI agents into daily automated briefings.
- Engineered a governance layer of seven risk checks plus a portfolio drawdown circuit breaker; medium-confidence signals escalate to human approval over Telegram and expire unanswered.
- Trained an XGBoost / RandomForest / LogisticRegression ensemble with walk-forward validation, a purged 21-day gap and Platt calibration; validated by 505 automated tests across 17 suites.
- Runs on Alpaca paper trading and deliberately stays there: a GO/NO-GO gate blocks live capital until the model clears 55 % accuracy (currently 52.9 %).

## EDUCATION

**B.S. in Systems and Software Engineering** Feb 2023 – Present | Pontificia Universidad Bolivariana

- Final-semester student. Advanced coursework in software architecture, system design and industry best practices.
- AWS Academy Cloud Foundations coursework — labs 1–6 completed: IAM, VPC, EC2, EBS, Auto Scaling and Load Balancing, with Terraform IaC implementations (2026).

**Scrum Master (academic project)** — *High-Availability Infrastructure* Jun 2025 – Nov 2025 | Pontificia Universidad Bolivariana

- Ran the full Scrum cycle: sprint planning, daily stand-ups, retrospectives and stakeholder demos.
- Contributed to the design of a high-availability infrastructure sized for about 100,000 concurrent users with a sub-500 ms latency objective (design targets, not measured outcomes), with hybrid SQL / NoSQL / in-memory data.

## CERTIFICATIONS

All credentials are publicly verifiable; full list with links at linkedin.com/in/johan-rodriguez-97bb29323.

- **Google Cloud** (2026) — Secure Lakehouse Data • Protect Sensitive Data with Data Loss Prevention • Create ML Models with BigQuery ML • Use APIs to Work with Cloud Storage • Orchestrate Multi-agent Workflows with Gemini Enterprise • Create Your First Gemini Enterprise Application • Build a Smart Cloud Application with Vibe Coding and MCP • Google DeepMind: Train A Small Language Model
- **Cisco Networking Academy** (2026) — Introduction to Cybersecurity
- **Hugging Face** (2026) — AI Agents Course
- **freeCodeCamp** (2026) — Machine Learning with Python • Data Analysis with Python • Scientific Computing with Python • Data Visualization • JavaScript Algorithms and Data Structures
- **EF SET** (2026) — English Certificate, CEFR C2 (Proficient), 74/100

## LANGUAGES

Spanish — Native | English — C2 Proficient (EF SET, verified)